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Trust Ecosystems in AI Governance

Building the Infrastructure Beneath the Rules

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Abstract

Every significant AI governance framework produced in the last five years — the EU AI Act, the NIST AI Risk Management Framework, ISO/IEC 42001, the G7 Hiroshima Process principles — calls for "trustworthy AI." None of them specifies what trust infrastructure is supposed to make that trustworthiness real. This paper argues that this omission is not a drafting detail left to implementers. It is the central problem in AI governance, and it is actively causing harm.

We develop a working theory of the trust ecosystem as the appropriate unit of analysis for AI governance: a structured arrangement comprising attestation infrastructure, verification mechanisms, shared interpretive frameworks, and accountability channels for when trust fails. Against this framework, we assess where current instruments succeed, where they fall short, and why those gaps are not accidental — they are the product of identifiable incentive problems, sovereignty concerns, and coordination failures that rules alone cannot resolve.

The paper then asks what a functional trust ecosystem would actually look like for AI systems operating across the EU-US-UK triangle, drawing on analogies from financial clearing systems and ICAO's aviation safety regime while being honest about where those analogies break. We close with four specific, actionable recommendations directed at regulators, standards bodies, and AI developers. The core argument is that trust is not a property of AI systems — it is a property of the ecosystems within which those systems are embedded. Governance that treats trustworthiness as an engineering problem will keep failing. The work ahead is institutional, relational, and infrastructural.

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1. The Governance Gap That Isn't Being Named

There is a phrase that appears in nearly every major AI governance document produced in the last five years. It appears in the preamble of the EU AI Act, in the mission statement of the NIST AI Risk Management Framework, in the Hiroshima Process principles issued by the G7, in the ISO/IEC 42001 standard for AI management systems, in national strategies from Singapore to Canada to the United Kingdom. The phrase is "trustworthy AI." And in almost none of these documents is it accompanied by a serious account of what trust infrastructure is supposed to produce that trustworthiness — who attests to what, how those attestations are checked, how trust signals propagate across organizational and jurisdictional boundaries, and what happens when the whole arrangement fails.

This is not a small oversight. It is the central problem in AI governance right now, and ignoring it is costing us. Governance frameworks can be beautifully structured, technically sophisticated, and politically hard-won, and they can still fail to produce trustworthy AI in practice if the infrastructure beneath the rules isn't there. That infrastructure — the attestation systems, the verification mechanisms, the shared standards that let different actors read each other's claims, the accountability channels — is what we mean by a trust ecosystem. And we do not yet have one. What we have instead is a collection of governance instruments that each invoke trust as a goal without specifying the social and institutional machinery required to achieve it.

1.1 What the Frameworks Actually Say

Start with the EU AI Act, which is the most legally forceful of the instruments. Regulation (EU) 2024/1689, adopted in June 2024, is built around a risk classification system and a set of conformity assessment requirements for high-risk AI systems. Its preamble makes clear that the regulation is motivated by "the need to build trust" and commits the EU to being "a global leader in the development of secure, trustworthy and ethical AI." The regulation requires that high-risk AI systems undergo conformity assessments, that providers maintain

technical documentation, that deployers register systems in a public database. These are real requirements with real teeth — fines can reach seven percent of global annual turnover for the most serious violations. But the regulation does not specify what infrastructure should exist to let a healthcare organization in Poland verify the conformity claims of an AI system developed in Germany and deployed across the MyHealth@EU framework. The conformity assessment regime is largely national in its enforcement architecture, and mutual recognition of assessments across third countries remains deeply underspecified.[1]

The NIST AI Risk Management Framework, released in January 2023, is different in character — it is explicitly voluntary, explicitly non-prescriptive, and designed to be adaptable across organizations of all sizes and sectors. NIST has been clear that the AI RMF is intended to "improve the ability to incorporate trustworthiness considerations into the design, development, use, and evaluation of AI products, services, and systems." [2] This is admirable as a statement of intent, and the framework's four core functions — Govern, Map, Measure, Manage — represent a genuine contribution to the field. But voluntary and non-interoperable by design means that two organizations that both "implement" the NIST AI RMF may be doing very different things and have no agreed mechanism for comparing or verifying each other's practices. There is nothing in the framework that constitutes attestation infrastructure, and NIST does not claim otherwise.

ISO/IEC 42001, published in late 2023, attempts to fill this gap with a certifiable management system standard — the first internationally agreed framework of its kind for AI governance. It establishes requirements for an AI management system, including policies, risk assessments, impact assessments, and a set of controls organized into nine control objectives.[3] Organizations can be certified against it by accredited certification bodies, and those certifications are technically comparable. This is progress. But as we will argue in Section 5, the standard is pitched at a level of abstraction that makes certified audits difficult to compare in practice, and it says very little about what happens when certified AI systems cross organizational or jurisdictional boundaries.

The G7 Hiroshima Process principles, issued in October 2023, are explicitly non-binding and addressed primarily to organizations developing advanced AI systems. They call for risk identification and mitigation, transparency, information sharing on incidents, and the development of technical standards. The principles are useful political signaling, but they explicitly acknowledge their own non-exhaustive and non-binding character, and they contain no enforcement mechanism of any kind.[4]

1.2 Why This Omission Is Not Benign

One might respond that these frameworks are doing real work — establishing legal requirements, promoting good practices, building a shared vocabulary — and that the absence of full trust infrastructure is a problem of implementation maturity, not fundamental design failure. We think this response is wrong, and here is why: the absence of trust

infrastructure does not leave a neutral gap. It leaves a vacuum that gets filled in costly and often harmful ways.

When there is no shared attestation infrastructure, organizations default to proprietary due diligence — every hospital system that wants to procure an AI diagnostic tool conducts its own vendor evaluation, with no agreed methodology, no shared evidence standards, and no way to verify the vendor's claims about training data or model behavior. This is expensive, it produces highly variable outcomes, and it systematically advantages large institutions over small ones. When there is no cross-border trust propagation mechanism, AI systems that have been assessed and approved in one jurisdiction face redundant, incommensurable assessments in every market they enter — creating enormous compliance burdens while providing little additional safety assurance, because the assessments don't share evidence or standards. When there are no accountability channels specified at the ecosystem level, harm that crosses organizational boundaries — a model failure that cascades through a supply chain, a bias that emerges in deployment at scale — falls into a liability vacuum that is resolved slowly, expensively, and inconsistently through litigation.

⚠ The Core Problem

Trust infrastructure is not the implementation detail that comes after the governance rules are written. It is the prerequisite that determines whether the rules can work at all. When governance frameworks call for "trustworthy AI" without specifying the trust infrastructure that produces trustworthiness, they are not leaving something to be worked out later — they are building on air.

There are concrete examples of this failure already visible in 2026. Healthcare AI systems cleared under FDA's predetermined change control plan (PCCP) pathways are operating in US health systems while their EU equivalents are still waiting for harmonized conformity assessment guidance under the AI Act's high-risk provisions, which apply to medical AI from August 2026.[5] The same underlying model architecture — perhaps the same weights — may be simultaneously "trustworthy" under one regulatory framework and awaiting certification under another, not because the systems are different but because the trust ecosystems are absent or incompatible. Cross-border data flows that underpin AI training and deployment are governed by GDPR adequacy decisions on the European side and a patchwork of state-level privacy laws and federal sectoral rules on the American side, with no agreed mechanism for assessing whether data governance standards meet the threshold for trust.

The point is not that current frameworks are useless. They are doing real work. The point is that they are doing that work without the structural support that would let them function as a coherent trust ecosystem rather than a collection of parallel, partially overlapping regimes. Naming this gap clearly is the precondition for closing it.

2. What Trust Actually Means in Sociotechnical Systems

Before we can design trust infrastructure, we need to be precise about what trust is. This matters more than it might seem, because governance practitioners tend to conflate three distinct things — trust, confidence, and compliance — and the conflation produces systematically bad policy. Treating them as equivalent leads to governance that optimizes for the wrong things.

2.1 Trust Is Not Confidence and Not Compliance

Annette Baier's foundational 1986 essay "Trust and Antitrust" remains the clearest starting point for a rigorous account.[6] Baier distinguishes trust from mere reliance. We rely on predictable mechanisms all the time — the timer on a dishwasher, the gravity that keeps our feet on the floor — without trusting them in any philosophically meaningful sense. Trust, on Baier's account, involves reliance on another's good will: a truster's willingness to make themselves vulnerable to a trustee in a domain, based on some belief about how the trustee will exercise their discretion when the truster is not watching. The key features of this account are the truster, the trustee, the domain of trust, and the basis for that trust. Crucially, trust involves vulnerability to not-yet-noticed harm or disguised ill will — it is not simply a bet on predictable behavior but an exposure to the possibility of betrayal.

Confidence, by contrast, is a probabilistic estimate. We are confident that a bridge will hold, a drug will work within its studied parameters, a statistical model will produce predictions within some error bound. Confidence is calibrated by evidence and can be appropriate even in systems that have no capacity for good will, because it does not invoke that concept at all. Compliance is different again — it is the state of having met a specified set of requirements. A system can be compliant with every applicable standard and still fail to be trustworthy, because compliance is assessed at a point in time against a specification, while trust is a dynamic relationship that depends on ongoing performance in conditions that specifications never fully anticipate.

Philip Pettit's work on trust in institutional contexts is useful here.[7] Pettit argues that certain forms of trust are not simply interpersonal but are constituted and maintained by institutional arrangements — by the expectation that actors are embedded in accountability structures that make betrayal costly and detectable. On this account, what institutions do when they work well is not replace trust with regulation, but rather create the conditions under which trust can operate safely. This is exactly right for AI governance: the goal of governance infrastructure is not to eliminate the need for trust but to give trust a viable structural foundation.



Key Distinction: Trust vs. Confidence vs. Compliance

Trust involves a truster making themselves vulnerable to a trustee in a specific domain, based on a belief about the trustee's good will. It is relational, dynamic, and involves exposure to potential betrayal.

Confidence is a probabilistic estimate of system behavior based on evidence. It requires no good will from the system — only predictability within known parameters.

Compliance is the static state of having met specified requirements at a point in time. A system can be fully compliant and untrustworthy — and often is, when specifications fail to anticipate real deployment conditions.

Good AI governance instruments should promote all three but must not confuse them. Most current frameworks are primarily compliance instruments, secondarily confidence-building tools, and almost never structural supports for trust in the deeper sense.

The conflation problem shows up acutely in how current frameworks handle the concept of "trustworthy AI." When the EU AI Act's high-level expert group articulated seven principles for trustworthy AI — human agency, technical robustness, privacy, transparency, fairness, societal well-being, and accountability — it was primarily describing desirable properties of AI systems rather than the structural conditions under which trust relationships between humans, institutions, and AI-embedded systems could form and persist.[8] These properties matter enormously. But describing properties of a system is not the same as designing the relational infrastructure within which trust can function. A painting can have all the properties of a masterpiece and still be a forgery; the property-list does not tell you how to build the authentication system.

2.2 The Distributed Trustee Problem

Classical accounts of trust work reasonably well when both truster and trustee are human agents with identifiable intentions and traceable discretion. In AI systems, the trustee is distributed across a supply chain — data providers, model developers, fine-tuners, cloud infrastructure providers, deploying organizations, system integrators — and none of these entities is fully in view of the end user. When a clinician uses an AI diagnostic tool, the object of their (implicit) trust is not a discrete agent with identifiable good will but a stack of technical and organizational decisions made by multiple parties, many of them upstream and invisible.

This is the distributed trustee problem, and it is more than a philosophical puzzle. It has direct governance implications. When trust breaks down — when the system produces a harmful recommendation, when bias becomes apparent, when a model behaves in deployment in ways that diverge from its stated training objectives — the distributed character of the trustee makes it extraordinarily difficult to allocate accountability. Each component of the stack may individually meet its applicable standards while the system-level behavior falls outside what any of them specified. This is not a hypothetical: it

is a characteristic feature of complex sociotechnical systems and a predictable consequence of supply chain AI deployment.

The philosophical literature on collective intentionality, and more specifically on institutional agency, is relevant here but under-applied in AI governance discourse. Margaret Gilbert's account of plural subjects and Raimo Tuomela's analysis of collective attitudes offer frameworks for thinking about how distributed actors can be held to shared commitments.[9] The practical translation is not straightforward, but the direction is clear: governing trust in AI systems requires governing the relationships among the distributed entities that collectively constitute the effective trustee. Current governance frameworks largely regulate individual actors at specific points in the supply chain; very few impose requirements on the relationships between those actors in ways that address the distributed trustee problem systematically.

2.3 Layers of Trust in AI Systems

Trust in AI systems operates at multiple layers simultaneously, and governance that ignores the layered structure will address some layers while leaving others unmanaged. We identify three primary layers, each of which has distinct infrastructure requirements.

The first is user-to-system trust: the relationship between an individual interacting with an AI-embedded product or service and the system itself. This is the layer that receives the most attention in current governance discourse — explainability requirements, transparency disclosures, human oversight mechanisms. It is important, but it is downstream of the other two layers. Users cannot make meaningful trust assessments of AI systems that are opaque at the institutional level.

The second is institution-to-institution trust: the relationship between the organizations involved in deploying, procuring, and overseeing AI systems. A hospital procuring an AI diagnostic tool is making a trust judgment about the developer's claims regarding model performance, training data quality, bias testing, and post-market monitoring. A regulator approving a high-risk AI system is making a trust judgment about the conformity assessment body that conducted the evaluation. These relationships are currently supported only by bilateral contracts, fragmented certification regimes, and legal liability frameworks that are slow and costly to invoke.

The third is cross-border trust propagation: the process by which trust signals generated in one jurisdiction or institutional context travel — or fail to travel — into others. This is the least developed and most consequential layer, and it is the subject of Section 4 of this paper. For now, it is enough to note that most current governance instruments are designed primarily for the first two layers, leaving the third almost entirely unaddressed.

 **Note on Baier and the AI Context**

Baier's account of trust as "reliance on another's good will" was developed in the context of interpersonal relationships, not technological systems. Applying it to AI requires care. The relevant extension is not that AI systems have good will in any literal sense, but that the human and institutional actors who design, train, deploy, and maintain those systems do — and that trust in the system is ultimately trust in those actors to have exercised their discretion appropriately in every decision that shaped the system's behavior. The distributed trustee problem is, at root, a problem about how to hold that distributed set of actors to the trust relationship that users and institutions are implicitly entering when they rely on AI systems.

3. Anatomy of a Trust Ecosystem

A trust ecosystem, as we use the term, is a structured set of arrangements — technical, institutional, legal, and procedural — that together enable trust to form, be verified, and be enforced across the actors and systems within a defined domain. The ecosystem metaphor is deliberate: trust ecosystems, like biological ecosystems, are characterized by interdependence among components, emergent properties that cannot be attributed to any single component, and the possibility of collapse when key components are absent or degraded. The metaphor is also a warning: ecosystems can appear healthy from a distance while critical linkages are failing, and partial or malformed ecosystems can be more dangerous than no ecosystem at all, because they create an illusion of coverage.

We identify four necessary components of a functional trust ecosystem. These are not stages of development or items on a checklist — they are structurally interdependent. Remove any one of them and the others cannot do their work.

3.1 Attestation Infrastructure

Attestation is the act of a recognized actor vouching that a specified claim holds, with defined scope, methodology, and liability. In trust ecosystems, attestation infrastructure is the bedrock: it is the layer that makes it possible for any actor to communicate verified claims about themselves or the systems they operate. Without it, every trust relationship has to be built from scratch through direct due diligence, which is expensive, inconsistent, and not scalable across complex multi-actor deployments.

There are mature attestation infrastructures in adjacent domains. Financial auditing — particularly the SOC 2 framework administered through the American Institute of Certified Public Accountants — provides a well-understood example. A SOC 2 Type II report attests, over a defined observation period, that an organization's controls for security, availability, processing integrity, confidentiality, and privacy meet specified trust service criteria. The attestation is produced by an independent auditor working to defined standards, carries defined liability, and is understood by a broad community of relying parties. It is not perfect

— as practitioners in the AI space have noted, SOC 2 was not designed to address AI-specific risks such as model bias, training data provenance, or runtime behavior drift — but it demonstrates what attestation infrastructure looks like when it works.[10]

In AI governance, attestation infrastructure is largely absent or immature. The EU AI Act's conformity assessment requirements for high-risk AI systems represent the most developed attempt: providers must either conduct internal assessments or use notified bodies, must produce technical documentation, and must register their systems in a Union database. But "notified body" status is granted by member states, the standards that notified bodies apply are still being harmonized, and there is no agreed methodology for what a conformity assessment actually demonstrates in terms that a relying party in a different jurisdiction can interpret. ISO/IEC 42001 certification comes closer to a globally recognized attestation mechanism — it uses accredited certification bodies applying a common standard — but its scope is the management system, not the AI system itself, and the granularity of its controls is insufficient to generate the specific, verifiable claims about model properties that trust relationships require.[11]

What functional attestation infrastructure for AI would require includes: recognized attestation bodies with defined competencies and accreditation; standardized attestation claims that specify scope, methodology, and validity period; liability frameworks that make false attestation costly; and machine-readable attestation formats that can be embedded in deployment documentation and propagated across supply chains. Some pieces of this are emerging — work on model cards, data sheets for datasets, and hardware-assisted attestation using trusted execution environments represents genuine technical progress — but they remain fragmented and not yet connected to governance infrastructure.[12]

3.2 Verification Mechanisms

Attestation establishes claims. Verification tests whether those claims hold. These are distinct functions that require distinct mechanisms, and a trust ecosystem that has one without the other is structurally incomplete. Attestation without verification is a market for credentials that no one can validate; verification without attestation produces site-specific knowledge that cannot accumulate or propagate.

Verification in the AI context is particularly challenging for reasons specific to the technology. First, AI systems are not static artifacts: a model that behaves in a specified way at the time of assessment may behave differently after fine-tuning, after deployment in a new environment, or after exposure to distribution shift in production data. This distinguishes AI systems from, say, medical devices or aircraft components, where the thing assessed and the thing deployed are largely the same thing. Second, many AI system properties that matter for trust — fairness across demographic subgroups, robustness to adversarial inputs, calibration of confidence estimates — require evaluation methods that are themselves contested and rapidly evolving. Third, the computational resources required

for thorough model evaluation are substantial, which means verification is expensive and available asymmetrically across actors of different sizes.

Current governance instruments address verification primarily through post-market monitoring requirements (the EU AI Act mandates this for high-risk systems) and incident reporting obligations (required under both the AI Act and, for frontier models, California's SB 53 from 2025). These are important but reactive: they identify problems after they occur rather than enabling relying parties to verify claims before they depend on them. What is needed in addition is a proactive technical verification infrastructure — standardized evaluation suites for common AI system properties, independent evaluation bodies with defined methodologies, and mechanisms for producing and communicating verification evidence to relying parties.

3.3 Shared Interpretive Frameworks

Even if attestation and verification are in place, a trust ecosystem requires that the claims produced can be read and understood by the actors who need to rely on them. This is the function of shared interpretive frameworks: the taxonomies, definitions, terminology, and conceptual structures that let different actors make sense of each other's assertions. Without them, attestation produces signals that relying parties cannot decode, and verification produces results that cannot be compared across contexts.

This problem is more acute in AI governance than in most adjacent domains, for two reasons. First, AI is technically complex and rapidly evolving, which means that the vocabulary needed to describe AI system properties precisely is itself contested and unstable. Terms like "bias," "fairness," "robustness," "explainability," and "alignment" mean different things in different technical communities and have different operational implications in different deployment contexts. A statement that a model has been "tested for bias" conveys almost no actionable information to a relying party without a shared understanding of what that testing involved, what definitions of bias it applied, and what population it was tested against. Second, AI governance is currently developing in parallel across multiple jurisdictions and standards bodies, each of which is developing its own terminology and categorization systems. The EU AI Act's risk categories, the NIST AI RMF's four-function structure, ISO/IEC 42001's control objectives, and the Hiroshima Process principles do not map cleanly onto one another, which means that compliance with one framework does not straightforwardly communicate anything about compliance with others.

Progress on shared interpretive frameworks requires active harmonization work at the standards level — something that is happening, but slowly. NIST has produced AI RMF crosswalks that attempt to map its framework onto ISO/IEC 42001 and other instruments.[2] The AI Office established under the EU AI Act is developing harmonized technical standards. But these efforts are largely driven by the same bodies that produced the frameworks themselves and have had limited participation from the technical communities doing the hardest verification work.

3.4 Accountability and Failure Channels

The fourth component of a trust ecosystem is the set of channels through which trust failures are detected, escalated, investigated, and remedied. Trust relationships that have no failure channels are not trust relationships — they are bets with no cover if they go wrong. The existence of credible accountability mechanisms is not just a remediation tool; it is a constitutive element of trust itself, because it structures the incentives of trustees and makes it rational for trusters to extend trust in the first place. This is Pettit's institutional trust logic applied directly: what institutions do is make betrayal of trust costly and detectable, which changes the trustee's incentives and makes trust rational rather than merely hopeful.[7]

In AI governance, accountability channels exist at the level of individual organizations (internal governance, board-level oversight) and at the level of national regulatory enforcement (fines under the EU AI Act, FTC enforcement in the US). What is almost entirely absent is accountability infrastructure at the cross-organizational and cross-border level — mechanisms for incident reporting that aggregate information across deployers, investigation authorities with jurisdiction over supply chains rather than individual actors, liability frameworks that can reach distributed trustees rather than only the nearest identifiable party.

The EU AI Act takes a step toward cross-organizational accountability with its incident reporting requirements: providers and deployers of high-risk AI systems must report serious incidents to national competent authorities, who feed information to the AI Office. But the regime does not yet include cross-border information sharing with non-EU regulators, and the definition of "serious incident" is calibrated to individual harms rather than systemic failures that accumulate gradually across many deployments. The aviation analogy is instructive here: the ICAO mandatory occurrence reporting system has been extraordinarily effective precisely because it creates a global incident database that reveals systemic patterns invisible to any single national regulator.[13] Nothing comparable exists for AI.

Defining the Trust Ecosystem

A **trust ecosystem** is a structured, interdependent arrangement of four components that together enable trust to form, be verified, and be enforced across the actors within a domain:

- 1. Attestation infrastructure** — who vouches for what, on what basis, with what liability, through what recognized process.
- 2. Verification mechanisms** — the technical and procedural means by which attestation claims are checked to determine whether they hold in practice.
- 3. Shared interpretive frameworks** — the standards, taxonomies, and definitional agreements that enable actors to read and compare each other's claims.

4. Accountability and failure channels — the mechanisms by which trust failures are detected, escalated, investigated, and remedied, creating the incentive structures that make trust rational rather than merely hopeful.

All four components are necessary. A governance arrangement that has three of the four is not three-quarters of a trust ecosystem — it is a structurally incomplete system whose gaps will be exploited by incentive problems.

4. The Propagation Problem

Of the four components of a trust ecosystem, the most underdeveloped — and the one that will matter most as AI deployment becomes genuinely global — is the infrastructure for trust propagation across jurisdictional and organizational boundaries. We can imagine a world in which individual jurisdictions develop reasonably functional trust ecosystems domestically. The EU has the strongest foundation for this, with mandatory conformity assessments, notified bodies, national competent authorities, and an AI Office providing central coordination. But a domestically functional trust ecosystem that cannot propagate its signals across borders is, at best, only half a solution in a world where AI supply chains, data flows, and deployment contexts routinely cross national lines.

4.1 How Trust Is Supposed to Travel

In other domains, trust propagation across jurisdictions works through a combination of mutual recognition agreements, harmonized standards, and interoperable oversight regimes. The logic is familiar: if jurisdiction A has assessed something using standards and methods that jurisdiction B judges equivalent to its own, then jurisdiction B can accept A's assessment without requiring a full independent re-evaluation. This is how much of international trade in regulated products works. The EU's mutual recognition agreements with trading partners allow products approved by EU notified bodies to enter certain markets without redundant certification. ICAO's Standards and Recommended Practices create a baseline safety regime that all 193 member states commit to upholding, which is why an aircraft certified under the FAA's type certification process can fly in European airspace under a bilateral safety agreement between the EU and the US.[13]

The financial sector offers another partial model. The Basel Accords create internationally harmonized capital adequacy standards that allow regulators in different jurisdictions to trust that banks operating cross-border are meeting a common minimum threshold. Clearing and settlement infrastructure — operated by entities like SWIFT and the major central counterparty clearinghouses — creates a technical layer that allows financial institutions across dozens of jurisdictions to transact with one another based on shared operational

standards and legal frameworks. Trust propagates through these systems because both the rules (Basel capital ratios) and the infrastructure (clearing systems, correspondent banking relationships) are interoperable.

None of this exists for AI at any meaningful scale. There are early movements toward bilateral AI governance cooperation — the EU-US Trade and Technology Council has working groups on AI standards, and the UK's AI Safety Institute has established partnerships with NIST and counterpart bodies in Japan and Singapore. But these are embryonic and have not produced interoperable attestation or mutual recognition of assessments. The current situation is one in which an AI system approved under one governance regime has no automatic standing under another, and the duplicative effort required to achieve compliance in multiple markets is borne almost entirely by AI developers, with large players able to absorb those costs and small players priced out of global markets.

4.2 Technical Barriers to Propagation

Trust propagation is difficult in AI for some specific technical reasons that do not apply with the same force to most other regulated sectors, and it is worth being precise about what these are.

The first is model opacity. Large language models and other foundation-model-class AI systems do not expose their decision logic in ways that can be straightforwardly assessed by an external party. An auditor evaluating a pharmaceutical drug can analyze its chemical composition and conduct controlled trials. An auditor evaluating a complex AI model can probe its behavior through test sets and red-teaming, but cannot directly inspect the internal processes that generate that behavior. This means that conformity assessments of AI systems are necessarily behavioral and inferential rather than compositional — they establish what the system does under specified conditions, not what it is. Behavioral assessments are dependent on the distribution of evaluation inputs, which means that a model that passes assessment under one evaluation regime may fail under another, not because it has changed but because the two regimes sample different regions of its input space.

The second is data provenance. Training data is the medium through which AI model behavior is shaped, and data provenance — the documented lineage of what data was used, how it was processed, and what rights and restrictions apply to it — is a prerequisite for meaningful trust assessment. Currently, data provenance documentation is inconsistent, non-standardized, and frequently proprietary. The GDPR creates strong data governance obligations on the European side, but there is no agreed framework for assessing whether data provenance documentation produced under one jurisdiction's standards constitutes adequate trust evidence under another's. Efforts to develop model cards and data sheets for datasets (work pioneered by researchers at Google and elsewhere and now referenced

in governance literature) represent important progress, but they remain voluntary, unverified, and not connected to any attestation infrastructure.[12]

The third is runtime behavior divergence. An AI system's behavior in production can diverge substantially from its behavior at the time of assessment, as a result of distribution shift (the statistical distribution of inputs in production differs from the training and evaluation distribution), model drift (the model's underlying patterns become less applicable over time), or integration effects (the system interacts with other components of the deployment environment in ways that produce emergent behaviors not present in isolation). This means that a point-in-time conformity assessment is a snapshot of a moving target, and that the trust it warrants degrades over time in ways that current governance instruments do not adequately account for. Aviation solves this problem through continuous airworthiness maintenance requirements — aircraft must be kept to certified standards throughout their operational life, not just at the time of initial certification. No equivalent regime exists for AI systems.

4.3 Institutional Barriers to Propagation

The technical barriers are real, but they are not the primary reason trust doesn't propagate in AI governance. Jurisdictions and institutions have navigated comparable technical challenges in other domains. The deeper barriers are institutional and political.

Liability asymmetry is one of the most significant. When jurisdiction A recognizes jurisdiction B's AI assessments, it is implicitly accepting some share of the liability consequences if those assessments turn out to be wrong. In a domain where AI harms are potentially large and poorly characterized, this is a significant exposure. No government wants to face domestic political consequences for a harm caused by an AI system that it accepted on the strength of another country's assessment. The aviation analogy breaks down here in an important way: when an aircraft certified by another authority crashes, the investigation is typically conducted by an international body (ICAO) under agreed protocols, and liability is allocated through a well-developed international legal framework. No such framework exists for AI.

Competitive disclosure aversion is a second major barrier. Much of the information required for meaningful AI trust assessment — training data composition, model architecture details, evaluation results across sensitive subpopulations — is competitively sensitive. AI developers have strong incentives to disclose as little as possible in conformity assessments and to resist governance regimes that would require disclosure to foreign regulators. Current frameworks manage this tension imperfectly: the EU AI Act protects trade secrets in technical documentation, but this protection makes it difficult for third-country regulators to assess whether documentation is substantively adequate or strategically incomplete.

Regulatory divergence driven by political economy is the third barrier. The EU, US, and UK are each developing AI governance frameworks that reflect different underlying political

choices — about the appropriate role of government, about the acceptable trade-off between innovation and precaution, about which harms are regulated and at what threshold. These divergences are not accidental; they reflect genuine differences in values and institutional cultures. Mutual recognition agreements work best when the underlying standards are substantively close. When they are genuinely different — as they are in AI, at least between the EU's rights-based precautionary approach and the US's innovation-first posture — mutual recognition requires one or both parties to accept standards they would not have chosen, which is politically costly and practically difficult.

4.4 Case Studies in Failed Trust Propagation

Two examples illustrate the propagation problem in concrete terms.

Healthcare AI across EU member states is a case in which propagation fails even within a single harmonized regulatory space. The MyHealth@EU framework is designed to enable cross-border exchange of health data, and the EU AI Act applies consistently across all member states. But in practice, a clinical AI system deployed in a German hospital that wishes to operate across the network in a French hospital faces a patchwork of national requirements — different notified bodies, different clinical validation expectations, different data governance interpretations — that the EU's formal harmonization has not yet resolved at the operational level. The infrastructure beneath the rules is still national even when the rules are European.[5] High-risk AI systems under the Act will be subject to mandatory conformity assessments from August 2026, but the technical standards that notified bodies apply are still being finalized, and the EU's AI Office is still developing harmonized quality-management and post-market monitoring guidance.

Large language models deployed across EU-UK-US jurisdictions present a starker case. A foundation model developed in the United States and deployed by a European enterprise may simultaneously be subject to: the EU AI Act's GPAI provisions (requiring transparency documentation and, for the most capable models, systemic risk management); the UK's principles-based AI governance approach (currently non-statutory and coordinated across sector regulators rather than through a dedicated AI Act); and a patchwork of US state-level and federal sectoral requirements. Data localization rules vary across all three jurisdictions. There is no agreed mechanism for the model provider to produce attestations that satisfy all three regulatory regimes through a common process, and no mechanism for any of the three regulators to rely on assessments produced for the others. Each compliance track is conducted in parallel, using different methodologies, producing incomparable outputs — and none of it produces a trust signal that propagates to the enterprise deployer's own regulators in a form they can use.

5. Where Current Frameworks Fall Short

We have been critical of current frameworks throughout this paper, but the critique requires precision. The major frameworks are not useless — they are doing important work within their scope. The question is not whether they contribute to AI governance, but whether together they constitute, or are building toward, a functional trust ecosystem. Our assessment is that they do not, and that understanding specifically what is missing from each framework is necessary before we can think clearly about what needs to be built.

5.1 Framework Assessment

The following table assesses each major framework against the four components of a trust ecosystem as defined in Section 3.

Framework	Attestation Infrastructure	Verification Mechanisms	Shared Interpretive Frameworks	Accountability & Failure Channels
EU AI Act (2024/1689)	Partial. Conformity assessments required for high-risk systems; notified body regime in development. Standards still being harmonized. Third-country recognition absent.	Partial. Post-market monitoring mandated; incident reporting required. No proactive independent verification infrastructure. Evaluation methods not standardized.	Moderate. Risk classification taxonomy is clear; detailed requirements exist. Crosswalk to ISO/IEC 42001 developing. GPAI tier thresholds defined but contested.	Moderate. National competent authorities and AI Office established from August 2025. Fines up to 7% global turnover. No cross-border incident database. Liability for third-country AI underdeveloped.
NIST AI RMF (2023)	Absent. Framework is voluntary and non-prescriptive; produces no attestation claims. Organizations may self-report RMF alignment	Weak. Playbook and profiles suggest verification practices but mandate nothing. Generative AI profile (2024) adds guidance	Strong. Four-function structure (Govern, Map, Measure, Manage) is widely adopted. Crosswalks to ISO/IEC 42001 and EU AI Act	Absent. Framework explicitly disclaims enforcement function. No incident reporting, no liability implications, no accountability

	with no independent verification.	but remains advisory.	produced. Most comprehensive US contribution to shared vocabulary.	mechanism of any kind.
ISO/IEC 42001 (2023)	Moderate. Certifiable management system standard with accredited certification bodies. 3-year certification cycle with annual surveillance. Scope is management system, not AI system properties directly.	Weak. Certification assesses management processes, not AI system behavior. No standardized AI property evaluation suite. Comparable audits difficult due to high abstraction level of controls.	Moderate. 38 controls across 9 objectives provide shared vocabulary. But controls are process-oriented rather than output-oriented, limiting comparability of certified organizations' AI systems.	Weak. Certification bodies can decertify; organizations face reputational consequences. But no mandatory incident reporting, no harm-specific liability, no cross-border investigation.
G7 Hiroshima Principles (2023)	Absent. Non-binding principles with no attestation mechanism.	Absent. No verification regime. Principles call for investment in safety research and red-teaming but do not establish verification infrastructure.	Weak. Principles build on OECD AI Principles but add limited definitional precision. No agreed methodology for what "safe, secure, and trustworthy" means operationally.	Absent. Explicitly non-binding. No enforcement, no liability, no investigation. Political signaling only.
UK AI Governance Approach (2025–26)	Absent at national level. Relies on sector regulators (FCA, CQC, Ofcom, ICO) applying existing	Partial. AI Safety Institute conducts pre-deployment evaluations of advanced AI models; partnerships with	Moderate. Principles-based approach enables flexibility but limits comparability. Alignment with	Partial. Sector regulators have enforcement powers under existing frameworks. No AI-specific

	frameworks to AI. AI Safety Institute produces evaluations of frontier models.	NIST, Japan AISI, Singapore. Coverage limited to frontier models.	EU AI Act sought but statutory framework still developing as of 2026.	incident reporting regime. Post-Brexit divergence from EU creates accountability gaps at border.
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Reading across this table, the pattern is clear. The frameworks collectively have reasonable coverage of shared interpretive frameworks — they are building shared vocabulary, however imperfectly. Verification and accountability are partial in the most developed frameworks and absent in the least. Attestation infrastructure — the component that must do the most work in trust propagation — is present only in embryonic form in the EU AI Act and ISO/IEC 42001, and entirely absent from the other frameworks. No framework addresses cross-border trust propagation at the infrastructure level at all.

5.2 Why the Gaps Are Structural, Not Accidental

The pattern of gaps is not random. It reflects identifiable structural forces that would produce the same result even if different people had written the frameworks with the best possible intentions. Understanding these structural forces is necessary because it shapes what remedies are realistic.

The first structural force is the incentive problem created by the competitive dynamics of AI development. Attestation infrastructure that works requires disclosure — of training data, model properties, evaluation results, deployment configurations. Disclosure is costly to the disclosing party in competitive terms, and the benefits of disclosure (enabling relying parties to make better trust judgments) accrue to the ecosystem as a whole rather than to the disclosing party. This is a classic collective action problem: every individual actor has incentives to seek the benefits of others' disclosure while minimizing their own, which produces a systematic underinvestment in the attestation infrastructure that everyone depends on. Solving this requires either mandatory disclosure regimes (which are politically difficult, especially across jurisdictions) or liability structures that make non-disclosure costly (which require legal frameworks that don't exist).

The second is the sovereignty concern that makes cross-border accountability infrastructure politically toxic. Any governance arrangement that would give a foreign regulator meaningful access to information about, or authority over, AI systems operating within national territory implicates sovereign prerogatives that governments are highly reluctant to cede. This is not irrational — it reflects real concerns about strategic technology dependence and political exposure. But it means that the cross-border accountability channels that a functional trust ecosystem requires can only be built on the basis of mutual agreement between jurisdictions that trust each other sufficiently to accept shared oversight, which is a trust problem at the inter-governmental level that cannot be solved by technical standards alone.

The third is the first-mover disadvantage problem in standards development. The organization or jurisdiction that develops a detailed attestation standard first bears significant costs — in research, in institutional investment, in the political effort required to get industry to accept it — while creating a public good that others can adopt freely. This produces systematic underinvestment in the hard standard-setting work and a tendency toward either very abstract standards (ISO/IEC 42001's level of abstraction is in part a product of this dynamic) or voluntary frameworks that ask nothing of their adopters (which is why NIST's AI RMF is deliberately non-prescriptive).

These structural forces are not insurmountable, but they can only be overcome by governance strategies that explicitly address them. Section 7 proposes recommendations designed to do exactly that.

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6. What Functional Trust Infrastructure Looks Like

Having established what is missing and why, we can now ask what functional AI trust infrastructure would actually look like — not as a utopian endpoint but as a plausible target state reachable by incremental, achievable steps. The question we want to answer is specific: what is the minimum viable trust ecosystem for AI systems operating across the EU-US-UK triangle, the three jurisdictions that together account for the largest share of advanced AI development and deployment?

6.1 Partial Models Worth Borrowing From

Two existing governance systems offer partial analogies that are worth taking seriously before we specify what AI needs.

ICAO's international aviation safety regime is the most-cited analogy in AI governance discourse, and for good reason. ICAO operates through a standard-setting body in which all 193 member states participate, producing Standards and Recommended Practices (SARPs) that are incorporated into national aviation law. National civil aviation authorities are responsible for implementation and enforcement within their jurisdictions, but mutual recognition agreements allow certificates of airworthiness issued by one major aviation authority (the FAA, EASA) to be accepted in other jurisdictions under bilateral agreements. The regime has an internationally interoperable incident reporting system that aggregates occurrence data across national authorities and produces systemic safety intelligence that no single authority could generate alone. The Trust Framework that ICAO has now developed for digital aviation data creates a federated identity and credentialing infrastructure for data exchange across borders.[13]

What works in the ICAO model: the combination of internationally agreed minimum standards (SARPs), bilateral mutual recognition agreements for major regulatory authorities, a shared incident reporting infrastructure, and a federated approach to oversight that preserves national sovereignty while enabling cross-border interoperability. What doesn't translate directly: aviation safety involves physical systems with deterministic behavior in defined operational envelopes. AI systems are probabilistic, adaptive, and context-dependent in ways that make the "certified to a fixed standard" logic fundamentally incomplete. Aviation also benefits from a highly concentrated industry structure (a small number of major airframe and engine manufacturers, a finite number of regulatory authorities that account for essentially all commercial aviation), which makes coordination tractable in ways that AI cannot match.

The financial clearing and settlement infrastructure offers a different kind of analogy. Systems like SWIFT, and the major central counterparty clearinghouses (CCPs), provide shared technical and legal infrastructure that allows financial institutions across dozens of jurisdictions to transact with each other on the basis of common operational standards and agreed rules for default and failure. The Basel Accords provide the underlying harmonized prudential standards that give regulators confidence that counterparties in other jurisdictions are maintaining minimum capital buffers. Critically, this infrastructure was built incrementally over decades, often in response to specific crises that demonstrated the cost of fragmentation. SWIFT emerged from a fragmented, inefficient telex-based system. The Basel framework emerged from the Herstatt Bank failure of 1974, which demonstrated how inadequate cross-border bank oversight could propagate failures internationally. There is a pattern here: trust infrastructure gets built when the cost of not having it becomes undeniable.[14]

The analogy breaks down in two important ways. Financial institutions operate within a common legal tradition of counterparty obligations and contract enforcement that has been developing for centuries; AI governance has no equivalent legal foundation for cross-border accountability. And financial transactions are discrete events with clear attribution; AI system outputs are continuous, contextual, and often not attributable to any specific identifiable decision by a specific identifiable actor.

6.2 A Minimal Viable Trust Ecosystem

With these analogies clearly scoped, what would a minimal viable trust ecosystem for AI governance across the EU-US-UK triangle actually require? We identify five necessary components, each of which is achievable with existing institutional building blocks and does not require the creation of a new global institution.

First, a federated AI system registry with interoperable identifiers. The EU AI Act already requires registration of high-risk AI systems in the EU database. The UK is developing parallel requirements. The US has no comparable federal mandate, but sector-specific agencies (FDA for medical AI, banking regulators for credit AI) maintain their own registries.

What is needed is not a unified global registry — that is not politically achievable — but interoperable identifiers that allow a system registered in one jurisdiction to be referenced reliably in others, and APIs that allow authorized parties to query registration status across jurisdictions. The ISO standard for Artificial Intelligence system identification (AI system lifecycle framework, ISO/IEC 23053) provides a starting point for identifier standardization.

Second, shared audit trail standards. AI systems operating in high-stakes domains should produce audit trails — logs of inputs, outputs, model versions, and decision configurations — in a format that is standardized enough to be read by auditors operating under different national frameworks. This does not require unified audit governance, only common technical standards for what audit logs contain and how they are structured. The financial sector has managed analogous standardization through SWIFT's messaging standards and the FIX protocol for trading data without requiring unified regulatory governance. A comparable effort for AI audit trails — ideally led by ISO with strong participation from NIST and ENISA — is technically feasible and would dramatically improve both domestic and cross-border verification.

Third, a liability allocation framework for cross-organizational AI supply chains. The distributed trustee problem cannot be solved by standards alone; it requires a legal mechanism for allocating liability across the actors in an AI supply chain when harms occur that no single actor can be held solely responsible for. The EU AI Act makes some progress here — it distinguishes providers from deployers and allocates different obligations to each — but it does not create a mechanism for allocating liability within the provider stack (between foundation model developers, fine-tuners, and integrators). A negotiated liability allocation framework, initially as a bilateral EU-UK instrument (given the closer regulatory relationship), would fill this gap and create a model for later extension to US-EU contexts.

Fourth, a cross-border AI incident reporting network. Following the ICAO occurrence reporting model, regulators in major AI governance jurisdictions should establish a shared incident reporting network with agreed definitions of what constitutes a reportable incident, minimum standards for incident reports, and an aggregation mechanism that produces systemic safety intelligence. The EU AI Office and the UK AI Safety Institute have both committed to transparency and information sharing in principle; the step required is converting those commitments into an operational data-sharing agreement with agreed technical standards. The US would need a single designated federal agency with authority to participate — something that could potentially be established under existing executive authority without new legislation in certain sectors.

Fifth, mutual recognition of conformity assessment bodies, beginning with ISO/IEC 42001 certifications and EU notified bodies. The EU already operates mutual recognition agreements for conformity assessments in many product categories. Extending this logic to AI governance, starting with the management-system layer (ISO/IEC 42001 certifications by accredited bodies) rather than the system-level conformity assessment layer (which is technically harder to harmonize), would create the first operational layer of cross-border

trust propagation and build the institutional relationships needed for more ambitious harmonization later.

A Note on Sequencing

These five components are not equally urgent, and the sequencing matters. Shared audit trail standards and a federated registry are relatively low political cost and high technical benefit; they can be pursued immediately through existing standards processes without requiring bilateral governmental negotiation. Liability allocation and mutual recognition are higher political cost and require sustained diplomatic engagement; they are medium-term objectives. The cross-border incident reporting network falls in between — technically straightforward, but requiring enough political trust between regulators to share sensitive information about AI failures in their jurisdictions. Building the first two components creates the institutional relationships and demonstrated good faith that make the others more achievable. The mistake would be to declare the whole architecture aspirational and defer action on any of it.

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7. Recommendations

The following four recommendations are directed at specific actors and address specific structural problems identified in this paper. We have written them as direct advice rather than hedged suggestions, because hedged suggestions are how governance discourse politely avoids commitment. Each recommendation identifies who needs to act, what the action is, and what a meaningful near-term step looks like.

Recommendation 1: The EU AI Office and NIST should jointly develop a binding cross-reference framework that maps EU AI Act conformity assessment categories to NIST AI RMF outcomes within eighteen months.

The absence of shared interpretive frameworks between the EU and US is the most tractable gap in the current ecosystem, because it is a coordination problem rather than a sovereignty conflict. The EU AI Office and NIST already have an institutional relationship through the EU-US Trade and Technology Council. The task is to convert that relationship into a concrete technical work product: a mapping document that specifies, for each category of high-risk AI system under the EU AI Act, what NIST AI RMF implementation outcomes would constitute evidence of meeting EU conformity requirements and vice versa. This is not a mutual recognition agreement — it does not require either jurisdiction to accept the other's assessments wholesale — but it creates a common reference frame that AI developers can use to design a single compliance process that generates evidence useful in both markets.

The near-term step is for the two bodies to establish a joint technical working group with a defined deliverable and a published timeline. NIST has already produced crosswalk documents for ISO/IEC 42001 and other frameworks; extending this methodology to the EU AI Act is within its existing operating model. The EU AI Office, operational from August 2025, has a mandate that includes "international alignment," which provides the institutional authority to engage.

Recommendation 2: Standards bodies — primarily ISO and NIST — should develop and publish an open technical standard for AI system audit logs within twelve months.

Audit trail standardization is the single intervention most likely to improve the verifiability of AI governance claims in the near term, at low political cost and high technical feasibility. The standard should specify: minimum fields required in audit logs for AI system inputs, outputs, model versions, configuration states, and operator interventions; data formats sufficient for interoperability across major cloud platforms; retention periods calibrated to the risk category of the AI system; and access controls that protect commercial confidentiality while enabling authorized regulatory inspection.

The near-term step is for NIST to issue a request for information on AI audit trail requirements, drawing on the experience of the healthcare AI and financial AI communities where audit requirements have been most developed. ISO/IEC JTC 1/SC 42 — the subcommittee responsible for AI standards — should take the NIST output as input to a fast-tracked standard development process targeting a Draft International Standard within eighteen months. The EU AI Act's technical documentation requirements provide a natural driver: developers who must meet those requirements will adopt whatever standard is available, creating a forcing mechanism for adoption.

Recommendation 3: The EU AI Office and the UK AI Safety Institute should conclude a bilateral AI incident reporting agreement, with agreed definitions and data-sharing protocols, by end of 2026.

The EU-UK relationship is the lowest-friction context for cross-border incident reporting because the two jurisdictions have a history of regulatory cooperation, share a legal tradition, and have regulatory bodies with established lines of communication from the post-Brexit transition period. An EU-UK incident reporting agreement would not require new primary legislation in either jurisdiction — it can be framed as a regulatory cooperation protocol under existing authorities. The agreement should specify: a common definition of "serious incident" in AI systems (distinct from each jurisdiction's domestic definition, harmonized at the more inclusive level); minimum data elements that incident reports must contain; a designated point of contact in each jurisdiction for cross-border notifications; a data-sharing arrangement that allows aggregated incident intelligence to be published without compromising individual operators' confidentiality; and a commitment to review the

protocol annually with a view to bringing the US, Japan, and Singapore into a multi-lateral framework.

The near-term step is for the heads of the EU AI Office and the UK AI Safety Institute to publicly commit to concluding this agreement by a specific date. Public commitment is the mechanism: both bodies have stated repeatedly that international information sharing is a priority; converting that stated priority into a signed protocol requires the accountability of a publicly announced deadline.

Recommendation 4: AI developers deploying high-stakes systems across multiple jurisdictions should immediately adopt supply-chain trust documentation as a standard practice, even in the absence of regulatory mandates.

This recommendation is directed at AI developers, not regulators, because the infrastructure gap is causing harm now and regulatory remedies will take years to fully develop. Supply-chain trust documentation means: for every AI system in high-stakes deployment, a maintained document that specifies the identity and certification status of every upstream provider (foundation model, data provider, fine-tuner, infrastructure provider); the attestation claims that each upstream provider makes and how those claims have been verified; the accountability mechanism in place with each upstream provider for incident notification and liability allocation; and the version control regime that governs model updates and the re-validation requirements that trigger on update. This is not currently required by any major governance framework, but nothing prevents it, and doing it creates competitive advantages that are likely to be recognized as procurement requirements emerge from public sector customers who are themselves subject to accountability obligations.

The near-term step is for major AI developers — and we name this as a challenge specifically to the frontier model providers who currently have the most market leverage and the least regulatory accountability — to publish their supply-chain trust documentation methodology and commit to independent verification of their documentation within twelve months. We are aware that this asks actors with weak incentives to do something costly. The answer is that the alternative — regulatory mandates designed by people who do not understand the technical constraints — is worse. The window for industry to shape this practice on its own terms is narrowing.

8. Conclusion

The central argument of this paper can be stated briefly, because it is not a complicated argument: it just happens to be one that governance discourse has systematically avoided making explicit. Trust is not a property of AI systems. It is a property of the ecosystems within which those systems are embedded — the attestation infrastructure, the verification mechanisms, the shared interpretive frameworks, and the accountability channels that together determine whether the actors who depend on AI systems have good reason to do so. Governance frameworks that treat trustworthiness as a feature to be engineered into a model, or as a checklist to be completed by a developer, are not describing a solution to the problem. They are describing one technical input to a social and institutional process that requires much more than technical inputs.

We have tried in this paper to be specific about what is missing and why. The major governance frameworks — the EU AI Act, the NIST AI RMF, ISO/IEC 42001, the Hiroshima Process — are real contributions, and we have credited them as such. But they are not sufficient, either individually or collectively, to constitute a trust ecosystem. The attestation layer is thin and fragmented. The verification layer is primarily reactive. The interpretive frameworks are better than they were two years ago but not yet interoperable across jurisdictions. And the accountability channels, particularly for cross-border and cross-organizational failures, are almost entirely absent.

The structural forces that produced these gaps — collective action problems in disclosure, sovereignty concerns about cross-border oversight, first-mover disadvantages in standards development — will not resolve themselves. They require deliberate governance responses targeted at their root causes. The recommendations in Section 7 are designed to do that: to create shared reference frames that reduce the cost of multi-jurisdictional compliance, to standardize the technical foundations of verifiability, to build cross-border incident intelligence, and to require AI developers to document the trust architecture of their supply chains.

There is a version of this argument that ends with a call for a new global institution — an IAEA for AI, or an ICAO for intelligent systems. We have deliberately not made that call, because we think the near-term policy space does not support it and because the institutional building blocks for a functional trust ecosystem exist already. The EU AI Office, NIST, the UK AISI, ISO/IEC JTC 1/SC 42, the bilateral regulatory relationships established through the EU-US Trade and Technology Council — these are not inadequate institutions waiting to be replaced. They are adequate institutions that have not yet been deployed in the right relationships with each other. Building a trust ecosystem is largely the work of connecting what already exists in ways that are currently not connected: creating the linkages, establishing the mutual references, agreeing the definitions, and building the accountability relationships that let the whole system function as more than the sum of its parts.

That work — institutional, relational, and infrastructural — is not glamorous. It does not produce landmark legislation or high-profile summits. It produces the infrastructure beneath

the rules: the unglamorous foundation without which the rules cannot do what they promise. It is, in our view, the most important work in AI governance right now, and it is the work that is most consistently being deferred in favor of the more visible and more politically rewarding activity of writing new principles.

We do not have the luxury of indefinite deferral. The AI systems are already deployed. The harms from inadequate trust infrastructure are already occurring. The window for building this ecosystem on a basis that reflects deliberate governance choice — rather than as a post-hoc response to a major failure — is closing. The argument of this paper is that we should use it.

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